

Question	Marking Guidance	Mark	Comments
10.1	1. Variation/differences due to mutation/s; 2. (Reference to) <u>allopatric</u> (speciation); 3. Smaller/different lakes have different environmental conditions OR Smaller/different lakes have different selection pressures; 4. Reproductive separation/isolation OR No gene flow OR Gene pools remain separate; 5. Different <u>alleles</u> passed on/selected OR Change in frequency of <u>allele/s</u> ; 6. Eventually different species/populations cannot breed to produce fertile offspring;	4 max	2. Ignore sympatric speciation. 3. Accept different populations for different lakes.
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10.2	1. Correct answer of 10/10.4 = 2 marks ;; 2. Working shows 14,112 = 1 mark OR 13.09/13.1 = 1 mark ;	2	1. Ignore any numbers after 10.4

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10.3	1. (Growth/increase of) algae/surface plants/algal bloom blocks light; 2. Reduced/no photosynthesis so (submerged) plants die; 3. Saprobiotic (microorganisms) aerobically respire OR Saprobiotic (microorganisms) use oxygen in respiration; 4. Less oxygen for fish to <u>respire</u> ;	4	3. Accept: Saprobiont/saprophyte/saprotroph 3. Neutral: decomposer
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10.4	1. Capture/collect sample, mark and release; 2. Ensure marking is not harmful (to fish) OR Ensure marking does not affect survival (of fish); 3. Allow (time for) fish to (randomly) distribute before collecting a second sample; 4. (Population =) number in first sample $\times$ number in second sample divided by number of marked fish in second sample/number recaptured;	4	2. Accept examples e.g., marking should not be toxic.
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10.5	1. Less chance of recapturing fish OR Unlikely fish distribute randomly/evenly;	1	Accept 'harder to capture marked fish' (recaptured fish) but ignore 'harder to capture fish'. Accept that fish may remain in one area. Accept fish may congregate.
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